

Image Generation Using Diffusion Models

Deep Learning Assignment

Name: Afzaal Kareem

Roll No: MSDS25062

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University: Information Technology University (ITU), Lahore

Abstract

This report presents the implementation of an image generation system based on Diffusion Models. The model was trained on a dataset of 100 images and learned to gradually remove noise from images through a denoising process. The objective was to understand the fundamentals of generative AI and evaluate the ability of diffusion models to synthesize new images.

Introduction

Diffusion Models are state-of-the-art generative models that create images by learning how to reverse a noise corruption process. During training, noise is progressively added to images, and the neural network learns to predict and remove this noise. During inference, the model starts from random noise and gradually reconstructs a meaningful image.

Dataset and Training Setup

Dataset Size: 100 images. Number of Batches: 13. Total Trainable Parameters: 5,330,819. The model was trained for 10 epochs. The loss consistently decreased, indicating successful learning and convergence.

Model Architecture

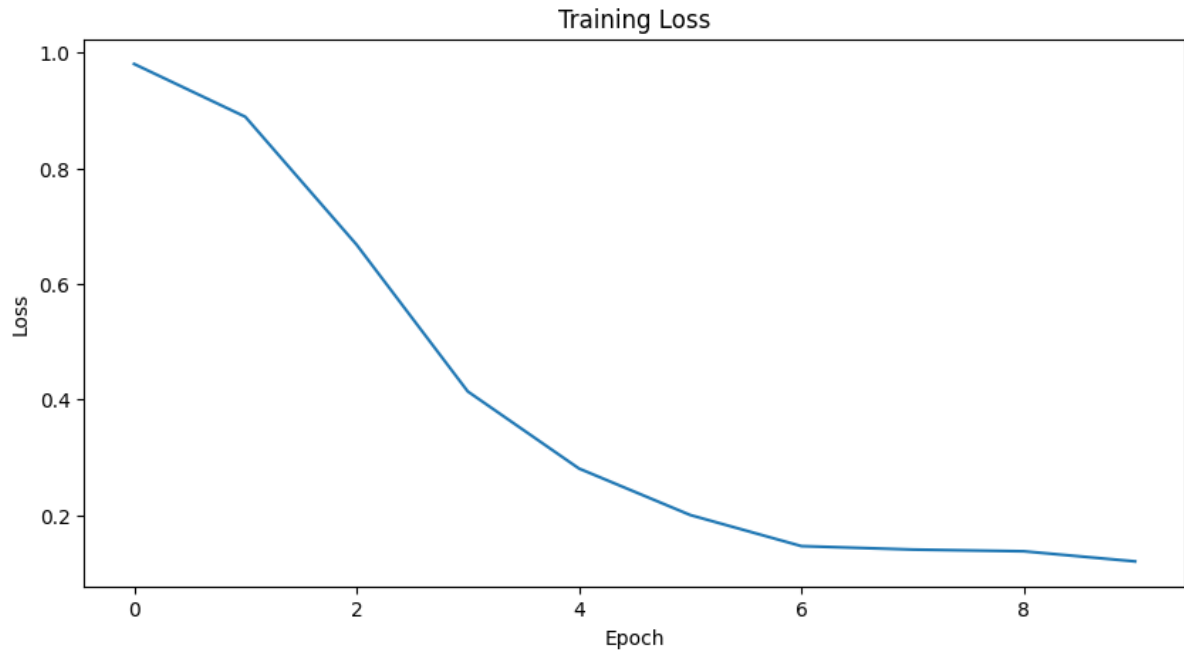
The diffusion framework consists of a forward diffusion process and a reverse denoising process. In the forward phase, Gaussian noise is added to training images over multiple timesteps. A deep neural network learns to estimate the noise present at each timestep. During image generation, the reverse process iteratively removes noise until a clean image is synthesized. This architecture enables the model to learn the underlying data distribution and generate realistic samples.

Working Process

1. Input images are loaded and normalized.
2. Noise is added at different timesteps.
3. The neural network predicts the added noise.
4. Prediction error is minimized using a loss function.
5. After training, random noise is provided as input.
6. The model iteratively removes noise to generate new images.

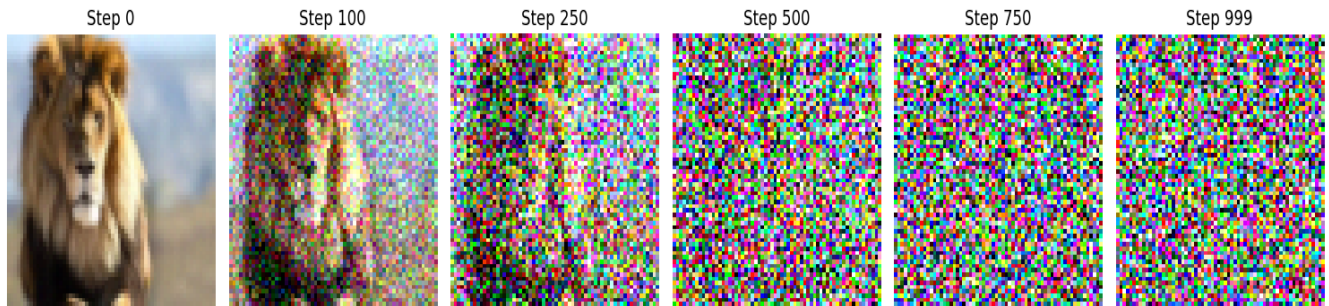
Training Results

Average training loss decreased from 0.9804 in Epoch 1 to 0.1199 in Epoch 10. This significant reduction demonstrates that the model successfully learned the denoising task and improved its prediction capability throughout training.



Diffusion Process Visualization

The visualization shows how the original image becomes increasingly noisy as the diffusion steps progress. At higher timesteps, the image is almost completely transformed into random noise. The model learns to reverse this process during generation.



Generated Samples and Discussion

The generated samples demonstrate that the model learned meaningful image structures from the training dataset. Although the generated outputs are not fully detailed due to the limited dataset size and short training duration, recognizable shapes and visual patterns emerge from random noise. With a larger dataset, more epochs, and additional computational resources, image quality can be significantly improved.

Generated Sample 1



Generated Sample 2



Generated Sample 3



Generated Sample 4



Conclusion

This assignment successfully implemented an image generation pipeline using diffusion models. The training loss consistently decreased across epochs, indicating effective learning. The inference stage confirmed that the trained model could generate new image samples from random noise. The results validate the capability of diffusion-based generative models and provide practical experience with modern deep learning techniques used in generative AI.